

TECHNOLOGY STACK

EDUGENIE – AI-POWERED LEARNING ASSISTANT

PROGRAMMING

- Python – primary backend and AI-module language.

BACKEND

- FastAPI – central web/API framework.
- Uvicorn – local ASGI server.
- Jinja2 – template rendering support.

AI

- Gemini 3.5 Flash – cloud-based generative AI for Q&A, quizzes, summaries and learning paths.
- LaMini-Flan-T5-783M – local model for concept explanations.

AI / ML SUPPORT

- Transformers
- PyTorch
- SentencePiece
- Accelerate

CONFIGURATION

- python-dotenv
- Environment variable configuration for Gemini API credentials.

FRONTEND

- HTML
- CSS

DEVELOPMENT

- Visual Studio Code
- Web browser
- Python virtual environment

CONNECTIVITY

Stable internet connectivity is required for Gemini API-based functions.